



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-2026
MONDAY TEST

CLASS: X
SUBJECT: CHEMISTRY

FULL MARKS: 40
DATE: 28/04/2025

General Instructions:

- The paper consists of three printed pages.
- Read the questions very carefully. Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

SECTION A (20 MARKS)

Attempt all questions

Question 1

Choose the correct answers from the options given below:

[8]

a) Some elements from the Periodic Table are given below:

Carbon, aluminium, fluorine, oxygen

How many element(s) has/ have electronegativity greater than that of nitrogen?

- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

b) An aqueous solution of lead nitrate forms a colourless solution on addition of:

- (i) Excess sodium hydroxide
- (ii) Drop wise sodium hydroxide
- (iii) Excess ammonium hydroxide
- (iv) Drop wise ammonium hydroxide

c) Assertion (A): Electronegativity of chlorine is higher than that of fluorine.

Reason (R) : Chlorine is placed just below fluorine in the same group in the periodic table.

- (i) Both A and R are true, and R is the correct explanation of A.
- (ii) Both A and R are true, and R is not the correct explanation of A.
- (iii) A is false but R is true.
- (iv) A is true but R is false.

d) If the empirical mass of the formula PQ_2 is 10, and the vapour density is 15, then the molecular formula will be:

- (i) PQ_2
- (ii) P_3Q_2
- (iii) P_6Q_3
- (iv) P_3Q_6

e) Choose the incorrect answer:

- (i) The covalent bonds are said to be polar when the shared pair of electrons are not at equal distance between the two bonded atoms.
- (ii) The bond formed between two atoms with slight difference in electronegativity is non polar.
- (iii) The bond formed between two atoms with low electronegativity difference is ionic.
- (iv) Polar covalent compounds ionise in water.

f) Assertion (A): Ions like Mg^{2+} , Na^+ , F^- and O^{2-} are isoelectronic species.

Reason (R) : All isoelectronic species contain different number of electrons.

- (i) A is true and R is the correct explanation of R.
- (ii) Both A and R are true, but R is not the correct explanation of A.
- (iii) Both A and R are false.
- (iv) R is false, but A is the true explanation.

- g) Elements A, B and C have electronic configuration as (2, 8, 3), (2, 8, 5) and (2, 8, 7) respectively. Which of the following holds true?
- B has the least electron affinity.
 - A has the highest electron affinity.
 - C has the highest electron affinity.
 - B has higher electronegativity than C.
- h) The compound which does not contain - one single, a double or a triple bond in its molecule.
- Nitrogen
 - Chlorine
 - Methane
 - Oxygen

Question 2

- a) The electronic configuration of three elements X, Y and Z are given below: [3]

Elements	Electronic configuration
X	2, 2
Y	2 8, 2
Z	2, 8, 8, 2

Arrange the elements X, Y and Z as per the instructions given below:

- Decreasing order of atomic size
 - Increasing order of metallic character
 - Increasing order of ionisation potential
- b) Draw the electron dot diagram for the following compounds given below: [3]
- Calcium oxide
 - Hydronium ion
 - Nitrogen molecule
- c) Calculate the empirical formula of the compound containing 37.6% sodium, 23.1% silicon, 39.3% oxygen. (Atomic mass: Na= 23, Si= 28, O=16) [3]
- d) The following table shows the tests a student performed on three different aqueous solutions X and Y. Based on the observations provided, identify the cations in X and Y. [2]

Chemical Test	Observation	Cation
(i) To a solution 'X' NH_4OH is added in minimum quantity first and then in excess	A gelatinous white ppt. is formed which dissolves in excess to form a clear solution	?
(ii) To a solution 'Y' NH_4OH is added in minimum quantity first and then in excess	A pale blue ppt. is formed which dissolves in excess to form a clear inky blue solution	?

- e) Define mole. [1]

SECTION B (20 MARKS)

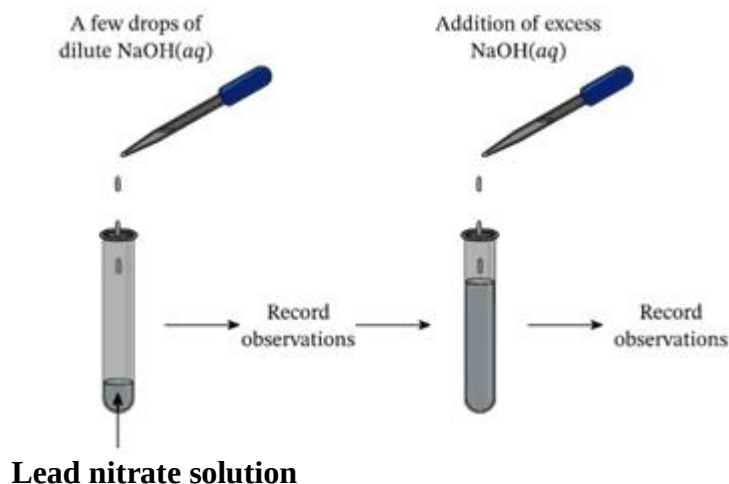
Attempt all questions

Question 3

- a) Calculate the percentage of nitrogen in aluminium nitride. [2]
[At. mass: Al = 27, N= 14, O=16]
- b) Inert gases have zero electron affinity. Account for the statement. [1]
- c) Calculate the volume of oxygen required to burn completely a mixture of 22.4dm^3 of methane and 11.2dm^3 of hydrogen. All volumes are measured at STP. ($1\text{ dm}^3=1\text{litre}$) [2]

Question 4

- a) Arun derived the molecular formula of a compound 'A' which has the following percentage composition by mass: C = 26.7%, O = 71.1%, H = 2.2%. He found the relative molecular mass of 'A' to be 90. Find the molecular formula derived by him.
[At. Mass: C = 12, H = 1, O = 16] [4]
- b) Study the diagram given below and write your observation. [1]



Question 5

- a) Two non metals combine with each other by sharing of electrons to form a compound 'X'. [3]
- (i) What type of chemical bond is present in 'X'?
- (ii) Will it be a good conductor of electricity or not? Give reason.
- b) Solve the following numericals: [2]
- (i) A gas of mass 32g has a volume of 20litre at STP. Calculate the molecular weight of the gas.
- (ii) Calculate the number of moles present in 160g of sodium hydroxide.
[At. Mass: Na = 23, O = 16]

Question 6:

Atomic number of element 'A' is 11 and it forms an ionic compound with 'B'. [5]

- a) Which of the following atomic numbers will match B?
- (i) 14 (ii) 10 (iii) 9
- b) Write the formula of the compound formed between A and B. Draw the electron dot structure of the compound formed between A and B.
- c) What is the name given to the members of the group to which B belongs to?
- d) Element ____ is an alkali metal.